

The Mathematics of the Modal Base Model:

ARIMA(0, 1, 1) $\equiv$ Simple Exponential Smoothing

Base forecaster in `mstl_multistep` — per-location, on the deseasonalised log series

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Abstract

On this data the per-location AutoARIMA most often selects order $(p, d, q) = (0, 1, 1)$ (the modal choice; $d=1$ in 93% of series, AR order usually 0). This note works through the mathematics of that model on a location's MSTL-deseasonalised log series D_t . ARIMA(0, 1, 1) is exactly **simple exponential smoothing (SES)**: a one-parameter EWMA whose point forecast is the flat last level, whose forecast standard deviation grows like $\sqrt{h}$, and whose one-step residual is white noise. Those three facts explain three things observed empirically in this model — the flat multi-step mean, the horizon-growing spread σ_h , and the near-noise residual the Random Forest is asked to correct.

1 Definition

Let B be the backshift operator, $B D_t = D_{t-1}$. Order $(p, d, q) = (0, 1, 1)$ means: difference once ($d = 1$), no autoregression ($p = 0$), and a first-order moving average on the difference ($q = 1$):

$$(1 - B) D_t = (1 + \theta B) \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2), \quad (1)$$

or written out,

$$D_t = D_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}. \quad (2)$$

There is a single dynamic parameter θ (plus the innovation variance σ^2), estimated by maximum likelihood per location. Invertibility requires $|\theta| < 1$.

2 It is simple exponential smoothing

Equation (1) is the reduced form of the *local-level* (SES) state-space model. Introduce a level ℓ_t ; the model is equivalent to the recursion

$$\hat{D}_{t+1|t} = \ell_t, \quad \ell_t = \alpha D_t + (1 - \alpha) \ell_{t-1}, \quad \boxed{\alpha = 1 + \theta} \quad (3)$$

so the level is an **exponentially-weighted moving average** of the observations,

$$\ell_t = \alpha \sum_{j \geq 0} (1 - \alpha)^j D_{t-j}. \quad (4)$$

The single estimated θ sets the memory: $\theta \rightarrow -1 \Rightarrow \alpha \rightarrow 0$ (very smooth, long memory); $\theta \rightarrow 0 \Rightarrow \alpha \rightarrow 1$ (a near random walk that just tracks the last value).

3 The forecast function is flat

With no AR term and one unit root, the h -step point forecast for *every* horizon equals the current level:

$$\hat{D}_{T+h|T} = \ell_T \quad \text{for all } h \geq 1. \quad (5)$$

This is the flat multi-step mean seen in practice (e.g. a forecast vector $[\mu_1, \mu_2, \mu_3]$ with all three entries equal). The ARIMA base therefore contributes essentially *no* horizon dynamics beyond the first step — it carries the smoothed last level forward — so any multi-step “signal” must come from the covariate (Random Forest) correction, not from ARIMA’s autoregression.

4 The predictive variance grows with horizon

ARIMA(0, 1, 1) is equivalent to ETS(A,N,N), whose h -step forecast-error variance is

$$\text{Var}(D_{T+h} - \hat{D}_{T+h|T}) = \sigma^2[1 + (h-1)\alpha^2], \quad \sigma_h = \sigma\sqrt{1 + (h-1)\alpha^2}. \quad (6)$$

Thus $\sigma_1 = \sigma$, $\sigma_2 = \sigma\sqrt{1 + \alpha^2}$, $\dots$: the variance is *linear* in h and the standard deviation grows like $\sqrt{h}$. This is exactly the σ_h the model reads off the 68% predictive interval and then widens with the variance head, $\sigma_{\text{eff}}^2 = \sigma_h^2 + \frac{1}{2}v(x)$.

5 What the Random Forest learns — and why the residual is noise

The one-step-ahead fitted value is $\hat{D}_{t|t-1} = \ell_{t-1}$, so the residual the RF is trained on is the model’s own innovation:

$$R_t = D_t - \hat{D}_{t|t-1} = D_t - \ell_{t-1} = \varepsilon_t, \quad (7)$$

and the level update is simply $\ell_t = \ell_{t-1} + \alpha R_t$. So R_t is **white noise by construction** once SES has absorbed the level. This is the mathematical reason the feature-to- R correlations are tiny ($|\rho| \leq 0.10$) and the RF correction is only a small nudge: the forest can recover only the *covariate-driven* structure that leaks into ε_t (climate / IRS effects SES cannot see), because there is no autoregressive structure left to exploit.

6 The drift variant

ARIMA(0, 1, 1) with drift adds a constant c to the differenced equation,

$$(1 - B)D_t = c + (1 + \theta B)\varepsilon_t \implies \hat{D}_{T+h|T} = \ell_T + ch, \quad (8)$$

turning the flat forecast (5) into a straight line of slope c (SES with a linear trend). It is useful only for the ~16% of series for which AutoARIMA includes a drift term.

Summary

The model’s typical base learner is one-parameter simple exponential smoothing: forecast = flat last level (5), interval widening like $\sqrt{h}$ (6), and a white-noise residual (7). Each of these is observable directly in the fitted model and accounts for the flat multi-step mean, the growing σ_h , and the near-noise RF target.